

Salman Haider

GIS Analyst | ArcGIS Pro, Spatial Analysis & Python Automation

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Summary

GIS professional with a Master of Geographic Information Systems from the University of Calgary and practical experience in ArcGIS Pro, spatial analysis, map production, remote sensing, spatial data QA/QC, Python/ArcPy workflow automation, and AI-assisted geoprocessing. I have delivered 105+ client-facing mapping projects and worked with cadastral, demographic, satellite imagery, transit, and routing datasets. My strength is turning spatial data into clean maps, reliable workflows, and clear outputs that help teams make better decisions.

Skills

GIS Platforms:

ArcGIS Pro • ArcGIS Online • ArcGIS Enterprise • ArcGIS Field Maps • ArcGIS Dashboards • ArcGIS StoryMaps • ArcGIS Experience Builder • QGIS • Global Mapper • Mango Maps • GeoServer • Leaflet • Mapbox GL JS • Mapbox Studio • MapTiler • Google Earth Engine • ENVI • ERDAS Imagine • Microsoft Office

Programming & Automation:

Python • ArcPy • SQL • JavaScript • HTML/CSS • Flask • GeoPandas • Shapely • GDAL • FME • PostGIS • PostgreSQL • MS SQL • Docker • Tippecanoe

Experience

Franchise Ready

Remote, Australia

GIS Coordinator

11/2021 - 08/2024

- Delivered **105+ franchise territory mapping** projects for clients across different industries, turning business briefings into GIS datasets, territory maps, reports, and web map deliverables.
- Worked from client briefs provided by the Director of Operations, reviewing each client's business model, competitors, target demographics, existing locations, sales patterns, and territory goals before starting the GIS work.
- Acted as the GIS point of contact during client follow-ups, explaining available datasets, mapping options, territory criteria, data limitations, and practical ways to fill gaps where suitable data was not readily available.
- Used ArcGIS Pro, ArcGIS Business Analyst, Australian Bureau of Statistics Census data, and demographic enrichment tools to build territory layers based on population, income, target demographics, competitor coverage, travel distance, and local market conditions.
- Created balanced franchise territories in ArcGIS Pro by combining client requirements with demographic patterns, competitor presence, existing sales information, POI access, and service-area coverage.
- Prepared client-ready cartographic products, territory summaries, project reports, ArcGIS Dashboards, and MangoMap web maps so clients could review territory boundaries and understand the analysis behind each recommendation.
- Automated repeatable project tasks with Python and ArcPy, including geocoding support, competitor and POI data preparation, spatial processing, map production steps, and quality checks to save time across recurring territory projects.

GreenAge Services

Islamabad, Pakistan

GIS Associate

04/2021 - 11/2021

- Created cadastral and land record maps using ArcMap for government land record digitization projects.
- Worked under a GIS Specialist on a government land record digitization project, converting old scanned paper maps of state land into structured digital GIS datasets.
- Used FME Workbench to convert AutoCAD plans and drawings into land parcel shapefiles for use in ArcMap and downstream land record mapping workflows.
- Created and maintained geodatabases for assigned project areas, digitizing state land parcels, boundaries, and related cadastral features with close attention to geometry and attribute accuracy.
- Collected ground control points with RTK GPS and supported georeferencing of scanned maps, satellite imagery, and drone imagery for accurate project basemaps.
- Wrote Python and OpenCV scripts to read scanned PDF/JPG map records, identify region information, and organize files into the correct folders for GIS associates working on assigned areas.
- Developed Python QA/QC scripts to check geodatabases and shapefiles for incorrect, missing, or misspelled attribute values before final mapping use.
- Prepared GIS databases, map layouts, basemaps, and supporting spatial datasets for planning workflows, internal review, and project communication.

Education

 **University of Calgary** Calgary, Canada
Master's degree, Geographic Information Systems 08/2024 - 05/2026

- Graduate training in spatial analysis, remote sensing, cartography, spatial databases, urban GIS, web GIS, GIS project design, and Python-based GIS automation.
- Worked with ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, enterprise geodatabases, data versioning, Experience Builder, StoryMaps, advanced geoprocessing, and projection workflows.

 **Arid Agriculture University** Rawalpindi, Pakistan
Bachelor of Science - Geoinformatics and Earth Observation 09/2016 - 10/2020

- Undergraduate training in GIS, remote sensing, cartography, photogrammetry, geodatabases, spatial analysis, surveying, web GIS, and image processing.
- Final year project used GIS and remote sensing methods to identify suitable rainwater harvesting sites in the Potohar region.

Academic Projects

Geo Copilot: Natural Language GIS Tool for ArcGIS Pro

github.com/salman12haider13/arcgis-geo-copilot-llm

ArcGIS Pro | ArcPy | Python | Large Language Models | Geoprocessing | MGIS Capstone

- Developed a custom ArcGIS Pro script tool (Geo Copilot) that enables users to perform geoprocessing tasks using natural language queries, integrating **Large Language Models (GPT-5 and Llama 3.1)** with ArcPy workflows.
- Designed structured reasoning frameworks to translate user intent into executable spatial workflows.
- Evaluated model performance across 36 geospatial test scenarios, achieving up to **91.7%** function selection accuracy and **83.3%** parameter accuracy with GPT-based workflows, and analyzed failure cases related to projection conflicts and multi-step reasoning.

Transit-to-Table: Calgary Fresh Food Access

ArcGIS Pro | GTFS | Network Analyst | OD Cost Matrix | Urban GIS | Accessibility Analysis | MGIS Project

- Built a GTFS-based multimodal transit network using Calgary Transit schedule data, street centreline data, walking connections, and ArcGIS Public Transit Data Model workflows.
- Modelled dissemination-area-level access from residential origin points to fresh food destinations using Network Analyst and a 45-minute weekday OD cost matrix.
- Combined OSM grocery and supermarket points with Calgary food hub locations to test how well transit connects neighbourhoods to fresh food options.
- Found **12 dissemination areas** with no transit-based access and **32 areas with poor access**, representing **33,212 residents** and showing uneven food accessibility across Calgary.

Ord's Kangaroo Rat Habitat Suitability Model

ArcGIS Pro | Suitability Modelling | Raster Analysis | Aspect Analysis | Habitat Mapping | MGIS Project

- Built an ArcGIS Pro habitat suitability model using **Model Builder** for Ord's Kangaroo Rat using DEM, active sand dunes, ecological range sites, unpaved roads, fireguards, and known species occurrence data from CFB Suffield.
- Prepared the analysis datasets by clipping, projecting, and standardizing DEM, road, fireguard, sand dune, and species occurrence layers to a common study area.
- Created the primary habitat model by identifying areas near active sand dunes, extracting dune tops with DEM focal statistics, and selecting south to southwest-facing dune aspects linked to suitable habitat.
- Developed the secondary habitat model using sandy road and fireguard conditions, dune proximity, and road orientation, then combined final rasters using a 70 percent primary habitat and 30 percent secondary habitat weighting.

Brazeau Dam Land Cover Classification and NDVI Time-Series Analysis

Google Earth Engine | JavaScript | ArcGIS Pro | Landsat 5, 7 and 8 | Image Classification | NDVI | MGIS Project

- Built a 4,567 square kilometre remote sensing study area around Brazeau Dam in Google Earth Engine using Landsat 5, Landsat 7, and Landsat 8 imagery.
- Created unsupervised and supervised land cover classifications using **K-Means clustering**, **CART classification**, training samples, validation samples, class remapping, and confusion matrix accuracy assessment.
- Produced NDVI median composites for 1990, 2005, and 2020 to compare vegetation conditions across a long-term time series.
- **Achieved 71.875 percent overall classification accuracy** and found vegetation decline by 2005 followed by recovery trends by 2020.

Academic Projects

Namal Lake Desiccation Assessment, 2000 to 2024

Remote Sensing | Google Earth Engine | Python | ArcGIS Pro | ERA5-Land | MNDWI | NDVI | OLS Regression | MGIS Project

- Processed Landsat 7, Landsat 8, and Sentinel-2 imagery in Google Earth Engine to measure Namal Lake surface area change from 2000 to 2024.
- Used **MNDWI** for annual water extraction and NDVI to estimate agricultural expansion around the lake, then built annual median composites and environmental time-series layers.
- Integrated ERA5-Land climate variables including temperature, precipitation, watershed evaporation, and lake surface evaporation, then tested relationships using **standardized OLS regression** in Python.
- Found a statistically significant decline in lake surface area, with agricultural expansion emerging as the strongest negative predictor and an **adjusted R² of 0.594**.

LiDAR and Object-Based Conifer Tree Height Classification

Remote Sensing | ENVI 6.0 | ArcGIS Pro | LiDAR CHM | QuickBird Imagery | Object-Based Image Analysis | SVM | KNN | MGIS Project

- Classified land cover and coniferous tree heights in a Campbell River, British Columbia study area using ENVI 6.0, ArcGIS Pro, QuickBird imagery, and a 1 metre LiDAR Canopy Height Model.
- Used NIR, RGB, and NDVI bands for example-based feature extraction, image segmentation, and training sample creation for roads, large trees, small trees, and clear-cut areas.
- Compared **KNN** and **SVM** classification methods and selected SVM because it performed better for identifying smaller clear-cut patches.
- Created nine LiDAR-based tree height classes from 0 to 45 metres, finding that the 25 to 30 metre class covered the largest area at 307,565 square metres.

Logistics Routing MVP

github.com/salman12haider13/Logistics-Routing-MVP-Demo

PostGIS | pgRouting | PostgreSQL | Flask | Docker | Leaflet | OpenStreetMap | GeoJSON | MGIS Project

- Built a self-hosted web GIS routing prototype using OpenStreetMap road data, PostgreSQL, PostGIS, pgRouting, Flask, Docker, and Leaflet.
- Designed the full routing workflow from raw OSM PBF data to a routable PostGIS network, including road filtering, topology creation, source and target nodes, travel-time costs, one-way handling, and spatial indexing.
- Developed a Flask backend API that accepts start and end points, routing mode, blocked polygons, traffic-delay points, and route alternative settings.
- Created an interactive Leaflet routing console that returns GeoJSON route geometry, route summaries, distance, travel time, route steps, blocked-zone routing behaviour, and alternative route options.

GEOSS Remote Sensing MVP Workflow for Vegetation Risk and Property Maintenance

Remote Sensing | Python | QGIS | PostgreSQL | PostGIS | Mapbox | AWS | Segment Anything Model | MGIS Project

- Designed a remote sensing MVP workflow for aerial vegetation identification, property risk assessment, and property maintenance planning using a sampled Vancouver study area.
- Proposed a processing pipeline using high-resolution orthophotos, Microsoft building footprints, PostGIS storage, Mapbox visualization, Python, QGIS, AWS processing, and SAM-based segmentation.
- Designed separate workflows for building segmentation, tree canopy extraction, canopy proximity analysis, and moss detection using buffers, clipped imagery, NDVI logic, and GeoJSON outputs.
- Framed the workflow around practical use cases including tree failure risk, moss growth, property damage, insurance risk, and municipal planning support, without presenting it as a full production deployment.

Courses

FME Desktop: Data Translation for AEC

LinkedIn

Creating a Web Map in ArcGIS Pro

Esri

Terrain Analysis Using ArcGIS Pro

Esri

Telling Stories with GIS Maps

Esri

PostgreSQL: Advanced Queries

LinkedIn

Using Python for Automation

LinkedIn

Location-Enabling Data

Esri

Mapping Addresses and Places

Esri

Advanced Python: Working with Databases

LinkedIn